Solving a Research Problem in Mathematical Statistics with AI Assistance Slides

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Solving a Research Problem in Mathematical Statistics with AI Assistance

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Abstract

Over the last few months, AI models including large language models have improved greatly. There are now several documented examples where they have helped professional mathematical scientists prove new results, sometimes even helping resolve known open problems. In this short note, we add another example to the list, by documenting how we were able to solve a previously unsolved research problem in robust mathematical statistics with crucial help from GPT-5. Our problem concerns robust density estimation, where the observations are perturbed by Wasserstein-bounded contaminations. In a previous preprint (Chao and Dobriban, 2023, arxiv:2308.01853v2), we have obtained upper and lower bounds on the minimax optimal estimation error; which were, however, not sharp.

Starting in October 2025, making significant use of GPT-5 Pro, we were able to derive the minimax optimal error rate (reported in version 3 of the above arxiv preprint). GPT-5 provided crucial help along the way, including by suggesting calculations that we did not think of, and techniques that were not familiar to us, such as the dynamic Benamou-Brenier formulation, for key steps in the analysis. Working with GPT-5 took a few weeks of effort, and we estimate that it could have taken several months to get the same results otherwise. At the same time, there are still areas where working with GPT-5 was challenging: it sometimes provided incorrect references, and glossed over details that sometimes took days of work to fill in. We outline our workflow and steps taken to mitigate issues. Overall, our work can serve as additional documentation for a new age of human-AI collaborative work in mathematical science.

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Human-only (Feb. 2021 - Aug. 2023):

$$\max\left\{n^{-2s/(2s+1)}, \varepsilon^{2s/(s+1/2)}\right\} \lesssim \mathcal{M}(\varepsilon, s, 1, 2, 2) \lesssim \max\left\{n^{-2s/(2s+1)}, \varepsilon^{2s/(s+2)}\right\}.$$

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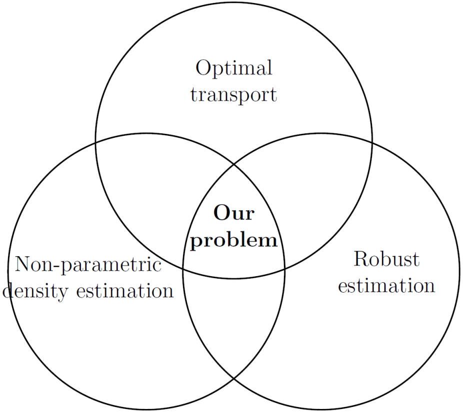
AI-assisted (Oct. - Nov. 2025):

$$\mathcal{M}(\varepsilon, s, p, q, r) \asymp \max\left\{n^{-\frac{2s}{2s+p}},\ \varepsilon^{\frac{2s}{s+1+p/r}}\right\}.$$

In particular, $\mathcal{M}(\varepsilon, s, 1, 2, 2) \asymp \max\left\{n^{-\frac{2s}{2s+1}},\ \varepsilon^{\frac{2s}{s+3/2}}\right\}$. Neither the upper nor the lower bounds were sharp.

Our Problem at the Intersection of Three Areas



Human-Only Lower Bound

In our **lower bound**, we have adapted the modulus of continuity approach, by starting with a normal density $f = \varphi_\sigma$ with sufficiently small variance $\sigma^2 > 0$, and perturbing it with a mean zero function to obtain a contaminated density $\tilde{f}$. Then, we upper bounded $W_{2,2}(f, \tilde{f}) \leqslant \varepsilon$ (via the Talagrand transportation cost inequality, see Talagrand (1996); Otto and Villani (2000)) and lower bounded $|f(x_0) - \tilde{f}(x_0)|$, which—by the modulus of continuity approach—lead to a lower bound on the estimation error.

AI-Suggested Lower Bound

GPT-5 Suggests Dynamic Benamou-Brenier for Improved Lower Bound (Extract)

User: [Prompt includes LaTeX draft of paper and a query: "Can you go through this proof and see if you can improve either the upper bound and the lower bound until they match?"]
GPT: [...]
Lemma [..] scales like $W_2^2 \lesssim D^2 \int g_+$, i.e. it is linear in the bump amplitude τ while the correct small–amplitude scaling for W_2 in one dimension is quadratic in τ.
[...] We state and prove the replacement for Lemma [..].

Lemma (Refined local W_2 bound via the dynamic formulation)

Let q be a probability density on $\mathbb{R}$ and let $I \subset \mathbb{R}$ be a finite interval. Let $g \in L^1(\mathbb{R})$ satisfy $\operatorname{supp}(g) \subset I$ *and* $\int g = 0$*. Write* $G(x) = \int_{-\infty}^{x} g(t)\,dt$*. Suppose there exists* $c_I > 0$ *such that* $q(x) \geq c_I$ *on* I *and* $|g|_\infty \leq c_I/2$*. Then* $q + g$ *is a density and*

$$W_2^2(q+g, q) \leq 2c_I \int_{\mathbb{R}} G(x)^2\,dx.$$

AI-Suggested Lower Bound (ctd)

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Proof.

Consider the path of densities $\rho_t = q + tg$ for $t \in [0,1]$. On I we have $\rho_t \geq q - |g|_\infty \geq c_I/2$, while off I one has $\rho_t = q > 0$. Define $v_t(x) = -G(x)/\rho_t(x)$. Then $\rho_t v_t = -G$ is independent of t, hence $\partial_x(\rho_t v_t) = -\partial_x G = -g$ and $\partial_t \rho_t = g$; thus $\partial_t \rho_t + \partial_x(\rho_t v_t) = 0$. Applying the Benamou–Brenier dynamic formulation gives

$$W_2^2(q+g,q) \leq \int_0^1 \int_{\mathbb{R}} |v_t(x)|^2 \rho_t(x)\, dx\, dt = \int_0^1 \int_{\mathbb{R}} G(x)^2\, dx\, dt = \int_{\mathbb{R}} G(x)^2\, dx.$$

Since $\rho_t(x) \geq c_I/2$ on I and $\rho_t = q$ off I, the bound sharpens to

$$W_2^2(q+g,q) \leq 2c_I \int_{\mathbb{R}} G(x)^2\, dx.$$

□